



# Ricardo Larraínzar Julve

## HARDWARE ENGINEER

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[LinkedIn](#) · [Technical Portfolio](#)

## ABOUT

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Hardware Engineer with experience in electronics for aerial platforms — from designing a power distribution board for drones (XRF) to analog design and data sensing on a race vehicle (Formula Student), with a placement at Aesculap (B. Braun Group) in Tuttlingen, Europe's largest medical device cluster. Skilled in LTSpice, PCB design, and bench validation.

## WORK EXPERIENCE

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### Junior Hardware & Robotics Engineer

XReality Factory (XRF), defense tech startup · Madrid, Spain · Sep 2025 – May 2026

Designed and developed energy systems for multirotor drones, including power architecture, current and voltage sensing, and PCB design.

### Biomechanical Engineering Intern

AESCU LAP, B. Braun Group · Tuttlingen, Germany · Jul 2023

Biomechanical engineering intern in the Biomechanics Lab, working on friction and wear testing, contributing to the development and analysis of tribological tests and diagnostics for medical devices in a regulated technical environment.

### Hardware Engineer

MAD FORMULA TEAM, Formula Student · Madrid, Spain · Oct 2022 – Sep 2023

Assisted in the design, fabrication, and programming of the PCB that collects sensor data from the MFT03 race car, as well as the wiring integrating those sensors.

## EDUCATION

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### Master's Degree in Industrial Engineering

UC3M, Madrid · 2024 – 2026

Thesis: *Design of an energy management system for UAV platforms*, developed in collaboration with XRF — redesigned a drone's power distribution board, reducing subsystem mass by 27% and shortening wiring.

### Bachelor's Degree in Industrial Electronics and Automation Engineering

UC3M, Madrid · 2020 – 2024

Thesis: *Design of a current-steering digital-to-analog converter* (10/10), simulated in LTSpice and integrated into a Sigma-Delta A/D converter.

Additional project: design of an analog theremin using optical distance sensors (group project).

## SKILLS

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- LTSpice
- Analog & Power Electronics Design
- MATLAB/Simulink
- PCB Design (KiCad)
- Instrumentation & Bench Validation
- Spanish (Native) · English (C1)